

PARMA NAND THAKUR

Pune, Maharashtra, India | +91 9955903610 | parmanandt109@gmail.com
[linkedin.com/in/parmanand](https://www.linkedin.com/in/parmanand) | github.com/parma-nand

Professional Summary

AI/ML Engineer with 4 years of experience at Capgemini, including 1.5+ years specializing in applied Generative AI: building production RAG systems, multi-agent LangGraph pipelines, and fine-tuning open-source LLMs. Strong foundation in backend engineering combined with hands-on expertise in retrieval-augmented generation, hybrid search, agentic workflows, and ML evaluation. Actively pursuing Applied AI/ML Engineer roles.

Technical Skills

- **Language:** Python
- **GenAI & Agents:** LangChain, LangGraph (Multi-Agent, StateGraph, ReAct), Hugging Face Transformers, LLM Fine-Tuning (LLaMA)
- **Retrieval & RAG:** ChromaDB, Qdrant, Hybrid Retrieval (BM25 + HNSW), CrossEncoder Reranking, RAGAS Evaluation
- **Machine Learning:** Classical Machine Learning, scikit-learn, XGBoost, Optuna, TensorFlow, Keras, PyTorch
- **Backend & Deployment:** FastAPI, Docker, Docker Compose, PostgreSQL, Spring Boot, Streamlit
- **Tools:** Git, GitHub, Postman, Playwright

Work Experience

Associate Consultant – AI/ML (GenAI) Engineer Jan 2025 – Present
Capgemini India Pvt. Ltd., Pune

- Architected and deployed a production-grade Retrieval-Augmented Generation (RAG) system using FastAPI, Streamlit, Qdrant, PostgreSQL, and Docker Compose for enterprise knowledge retrieval.
- Implemented hybrid retrieval combining BM25 sparse search with HNSW dense vector search to improve recall and precision across large document corpora.
- Integrated CrossEncoder-based reranking to refine retrieved context and improve response relevance for downstream LLM generation.
- Established an automated evaluation framework using RAGAS, achieving 1.0 faithfulness and 0.97 answer relevancy scores.
- Fine-tuned open-source LLMs (LLaMA) on local GPU infrastructure for domain-specific use cases, optimizing for latency and accuracy.
- Designed and maintained a structured prompt and system-prompt library, standardizing LLM behavior across multiple GenAI applications.
- Architected multi-agent workflows using LangGraph, implementing supervisor, hierarchical, and parallel agent patterns for complex reasoning tasks.
- Implemented Pydantic-based structured output schemas to ensure reliable, type-safe communication between agent nodes.
- Built and managed vector database infrastructure (ChromaDB, FAISS, Qdrant) supporting semantic search across large-scale document collections.
- Developed document ingestion pipelines for chunking, embedding generation, and indexing using sentence-transformers.
- Containerized and deployed AI microservices using Docker and Docker Compose across development, staging, and production environments.
- Collaborated with a 5-person GenAI team in Agile sprints, contributing to architecture design, code reviews, and client-facing technical demos.

Associate – Backend Support Engineer Jul 2022 – Dec 2024
Capgemini India Pvt. Ltd., Pune

- Developed and maintained backend services using Java and Spring Boot, supporting production systems with high-availability requirements.
- Diagnosed and resolved production incidents, contributing to improved system reliability and reduced downtime.
- Built and maintained REST APIs, collaborating with cross-functional teams in an Agile environment.

Projects

Production RAG System – Hybrid Retrieval & Evaluation [GitHub/RAG](#)

- End-to-end RAG pipeline (FastAPI, Streamlit, Qdrant, PostgreSQL, Docker Compose) combining BM25 sparse retrieval with HNSW dense retrieval and CrossEncoder reranking; evaluated with RAGAS (.9 faithfulness, 0.85 answer relevancy).

Document Intelligence: PDF Parsing, Chunking & Semantic Search [GitHub/NLP](#)

- Multi-column PDF text extraction (PyMuPDF, pdfplumber) with semantic chunking and embedding generation (sentence-transformers), indexed in FAISS/ChromaDB for semantic search.

Multi-Agent Job Application Automation (LangGraph) – In Progress [GitHub/Agentic Ai](#)

- Multi-agent system (Scout, Screener, Resume, Applier) built with LangGraph StateGraph, Pydantic structured outputs, Playwright, FastAPI, and ChromaDB for automated job search, matching, and application.

ML Pipeline: XGBoost + Optuna Hyperparameter Tuning [GitHub/XGBoost](#)

- Built a classification pipeline using XGBoost with Optuna-driven hyperparameter search. *[Add dataset/domain and metric, e.g., "Improved validation AUC from 0.81 to 0.89 on ..."]*

Voice Chat AI Assistant

[GitHub/Voice Chat](#)

- Voice-driven conversational AI using Whisper (STT) and TTS, with LangChain memory for multi-turn dialogue; deployed as a containerized FastAPI microservice supporting concurrent sessions with sub-2-second latency.

Education

National Institute of Technology (NIT) Jalandhar

Aug 2018 – May 2022

Bachelor of Technology (B.Tech)

Certifications

- Amazon Web Services (Coursera) – AWS Cloud Technical Essentials